

```

if self._waiter is not None:
    raise RuntimeError(
        f'{func_name}() called while another coroutine is '
        f'already waiting for incoming data')

assert not self._eof, '_wait_for_data after EOF'

# Waiting for data while paused will make deadlock, so prevent it.
# This is essential for readexactly(n) for case when n > self._limit.
if self._paused:
    self._paused = False
    self._transport.resume_reading()

self._waiter = self._loop.create_future()
try:
>     await self._waiter
E     RuntimeError: Task <Task pending name='Task-82' coro=<StreamReader.read()
running at
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>
cb=[_release_waiter(<Future pending...ask_wakeup()>)]() at /Users/akhilarya/UW Madison &
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait
_for.py:38]> got Future <Future pending> attached to a different loop

/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:
RuntimeError
_ ERROR at setup of TestMilestone1AcceptanceCriteria.test_AC4_dual_layer_retrieval _

self = <nexus.epistemic.graph_store.GraphStore object at 0x11fc91a50>

@asynccontextmanager
async def _session(self) -> AsyncGenerator[AsyncSession, None]:
    """Yield a fresh session, ensuring the driver is open."""
    if self._driver is None:
        raise RuntimeError("GraphStore not connected — call await store.connect() first")
    async with self._driver.session(database=self._database) as session:
>     yield session

nexus/epistemic/graph_store.py:130:
-----
nexus/epistemic/graph_store.py:165: in upsert_node
    result = await session.run(
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:338: in run
    await auto_result._run(
.venv/lib/python3.11/site-packages/neo4j/_async/work/result.py:240: in _run

```

```

    await self._attach()
.venv/lib/python3.11/site-packages/neo4j/_async/work/result.py:441: in _attach
    await self._connection.fetch_message()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:204: in inner
    await coroutine_func(*args, **kwargs)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:865: in fetch_message
    tag, fields = await self.inbox.pop()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:77: in pop
    await self._buffer_one_chunk()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:54: in _buffer_one_chunk
    await receive_into_buffer(self._socket, self._buffer, 2)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:340: in receive_into_buffer
    n = await sock.recv_into(
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:221: in
recv_into
    res = await self._wait_for_read(self._reader.read, nbytes)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:173: in
_wait_for_io
    return await wait_for(io_fut, timeout)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait_for.py:118: in wait_for
    return fut.result()
    ^^^^^^^^^^^^^^^^^
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694: in
read
    await self._wait_for_data('read')

```

```

-----

self = <StreamReader transport=<_SelectorSocketTransport fd=22>>
func_name = 'read'

```

```

async def _wait_for_data(self, func_name):
    """Wait until feed_data() or feed_eof() is called.

```

```

    If stream was paused, automatically resume it.
    """

```

```

    # StreamReader uses a future to link the protocol feed_data() method
    # to a read coroutine. Running two read coroutines at the same time
    # would have an unexpected behaviour. It would not possible to know
    # which coroutine would get the next data.

```

```

    if self._waiter is not None:

```

```

        raise RuntimeError(
            f'{func_name}() called while another coroutine is '

```

```

        f'already waiting for incoming data')

    assert not self._eof, '_wait_for_data after EOF'

    # Waiting for data while paused will make deadlock, so prevent it.
    # This is essential for readexactly(n) for case when n > self._limit.
    if self._paused:
        self._paused = False
        self._transport.resume_reading()

    self._waiter = self._loop.create_future()
    try:
        > await self._waiter
    E     RuntimeError: Task <Task pending name='Task-79' coro=<StreamReader.read()
    running at
    /Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>
    cb=[_release_waiter(<Future pending...ask_wakeup()>)]() at /Users/akhilarya/UW Madison &
    F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait
    _for.py:38]> got Future <Future pending> attached to a different loop

    /Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:
    RuntimeError

```

During handling of the above exception, another exception occurred:

```
self = <neo4j._async.work.session.AsyncSession object at 0x125a7ebd0>
```

```
@AsyncNonConcurrentMethodChecker._non_concurrent_method
```

```
async def close(self) -> None:
```

```
    """
```

```
    Close the session.
```

```
    This will release any borrowed resources, such as connections, and will
    roll back any outstanding transactions.
```

```
    """
```

```
    if self._closed:
```

```
        return
```

```
    if self._connection:
```

```
        if self._auto_result and self._state_failed is False:
```

```
            try:
```

```
                await self._auto_result.consume()
```

```
                await self._update_bookmark(self._auto_result._bookmark)
```

```
            except Exception:
```

```
                # TODO: Investigate potential non graceful close states
```

```

        self._auto_result = None
        self._state_failed = True

    if self._transaction:
        if self._transaction._closed() is False:
            # roll back the transaction if it is not closed
            await self._transaction._rollback()
        self._transaction = None

    try:
        if self._connection:
            await self._connection.send_all()
>         await self._connection.fetch_all()

.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:216:
-----
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:883: in fetch_all
    detail_delta, summary_delta = await self.fetch_message()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:865: in fetch_message
    tag, fields = await self.inbox.pop()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:77: in pop
    await self._buffer_one_chunk()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:54: in _buffer_one_chunk
    await receive_into_buffer(self._socket, self._buffer, 2)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:340: in receive_into_buffer
    n = await sock.recv_into(
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:221: in
recv_into
    res = await self._wait_for_read(self._reader.read, nbytes)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:173: in
_wait_for_io
    return await wait_for(io_fut, timeout)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait_for.py:118: in wait_for
    return fut.result()
    ^^^^^^^^^^^^^^^^^
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694: in
read
    await self._wait_for_data('read')
-----

self = <StreamReader transport=<_SelectorSocketTransport fd=22>>

```

```
func_name = 'read'
```

```
async def _wait_for_data(self, func_name):  
    """Wait until feed_data() or feed_eof() is called.
```

```
  
    If stream was paused, automatically resume it.  
    """
```

```
    # StreamReader uses a future to link the protocol feed_data() method  
    # to a read coroutine. Running two read coroutines at the same time  
    # would have an unexpected behaviour. It would not possible to know  
    # which coroutine would get the next data.
```

```
    if self._waiter is not None:  
        raise RuntimeError(  
            f'{func_name}() called while another coroutine is '  
            f'already waiting for incoming data')
```

```
    assert not self._eof, '_wait_for_data after EOF'
```

```
    # Waiting for data while paused will make deadlock, so prevent it.  
    # This is essential for readexactly(n) for case when n > self._limit.
```

```
    if self._paused:  
        self._paused = False  
        self._transport.resume_reading()
```

```
    self._waiter = self._loop.create_future()  
    try:
```

```
>         await self._waiter
```

```
E         RuntimeError: Task <Task pending name='Task-80' coro=<StreamReader.read()  
running at
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>  
cb=_release_waiter(<Future pending...ask_wakeup())>()) at /Users/akhilarya/UW Madison &  
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait  
_for.py:38]> got Future <Future pending> attached to a different loop
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:  
RuntimeError
```

During handling of the above exception, another exception occurred:

```
engine = <nexus.epistemic.engine.EpistemicEngine object at 0x11fcf7b10>
```

```
@pytest_asyncio.fixture(scope="session")  
async def seeded_engine(engine: EpistemicEngine):  
    """
```

Engine pre-loaded with a small synthetic infrastructure dataset.
Seeded ONCE for the session; all read-only tests share this state.

"""

```
nodes = [
    GraphNode(type=EntityType.SERVICE, name="api-gateway", namespace="k8s"),
    GraphNode(type=EntityType.SERVICE, name="auth-service", namespace="k8s"),
    GraphNode(type=EntityType.SERVICE, name="user-service", namespace="k8s"),
    GraphNode(type=EntityType.DATABASE, name="postgres-main", namespace="k8s"),
    GraphNode(type=EntityType.DATABASE, name="redis-cache", namespace="k8s"),
    GraphNode(type=EntityType.ENDPOINT, name="/api/v1/login", namespace="k8s"),
    GraphNode(type=EntityType.TABLE, name="users", namespace="k8s"),
    GraphNode(type=EntityType.TABLE, name="sessions", namespace="k8s"),
    GraphNode(type=EntityType.MODULE, name="auth.py", namespace="python"),
    GraphNode(type=EntityType.FUNCTION, name="verify_token",
namespace="python"),
]
saved_nodes = []
for node in nodes:
>     saved = await engine.add_node(node)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

tests/integration/test_milestone1.py:81:

```
-----
nexus/observability/tracing.py:148: in wrapper
    return await func(*args, **kwargs)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
nexus/epistemic/engine.py:247: in add_node
    return await self._graph.upsert_node(node)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
nexus/observability/tracing.py:148: in wrapper
    return await func(*args, **kwargs)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
nexus/epistemic/graph_store.py:164: in upsert_node
    async with self._session() as session:
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/contextlib.py:222: in
__aexit__
    await self.gen.athrow(typ, value, traceback)
nexus/epistemic/graph_store.py:129: in _session
    async with self._driver.session(database=self._database) as session:
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:122: in __aexit__
    await self.close()
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:227: in close
    await self._disconnect()
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:137: in _disconnect
```

```

return await super()._disconnect(sync=sync)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

.venv/lib/python3.11/site-packages/neo4j/_async/work/workspace.py:285: in _disconnect
    await self._pool.release(self._connection)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_pool.py:498: in release
    await connection.reset()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt5.py:443: in reset
    await self.fetch_all()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:883: in fetch_all
    detail_delta, summary_delta = await self.fetch_message()
    AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:865: in fetch_message
    tag, fields = await self.inbox.pop()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:77: in pop
    await self._buffer_one_chunk()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:54: in _buffer_one_chunk
    await receive_into_buffer(self._socket, self._buffer, 2)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:340: in receive_into_buffer
    n = await sock.recv_into(
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:221: in
recv_into
    res = await self._wait_for_read(self._reader.read, nbytes)
    AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:173: in
_wait_for_io
    return await wait_for(io_fut, timeout)
    AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait_for.py:118: in wait_for
    return fut.result()
    AAAAAAAAAAAAAA

/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694: in
read
    await self._wait_for_data('read')

```

```

-----

self = <StreamReader transport=<_SelectorSocketTransport fd=22>>
func_name = 'read'

```

```

async def _wait_for_data(self, func_name):
    """Wait until feed_data() or feed_eof() is called.

```

```

    If stream was paused, automatically resume it.
    """

```

```

    # StreamReader uses a future to link the protocol feed_data() method

```

```

# to a read coroutine. Running two read coroutines at the same time
# would have an unexpected behaviour. It would not possible to know
# which coroutine would get the next data.
if self._waiter is not None:
    raise RuntimeError(
        f'{func_name}() called while another coroutine is '
        f'already waiting for incoming data')

assert not self._eof, '_wait_for_data after EOF'

# Waiting for data while paused will make deadlock, so prevent it.
# This is essential for readexactly(n) for case when n > self._limit.
if self._paused:
    self._paused = False
    self._transport.resume_reading()

self._waiter = self._loop.create_future()
try:
>     await self._waiter
E     RuntimeError: Task <Task pending name='Task-82' coro=<StreamReader.read()
running at
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>
cb=[_release_waiter(<Future pendi...ask_wakeup()>)]() at /Users/akhilarya/UW Madison &
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait
_for.py:38]> got Future <Future pending> attached to a different loop

/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:
RuntimeError
===== FAILURES =====
_____ TestGraphStore.test_upsert_and_retrieve_node _____

self = <nexus.epistemic.graph_store.GraphStore object at 0x11fc91a50>

@asynccontextmanager
async def _session(self) -> AsyncGenerator[AsyncSession, None]:
    """Yield a fresh session, ensuring the driver is open."""
    if self._driver is None:
        raise RuntimeError("GraphStore not connected — call await store.connect() first")
    async with self._driver.session(database=self._database) as session:
>         yield session

nexus/epistemic/graph_store.py:130:
-----
nexus/epistemic/graph_store.py:165: in upsert_node

```



```

    result = await session.run(
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:338: in run
    await auto_result._run(
.venv/lib/python3.11/site-packages/neo4j/_async/work/result.py:240: in _run
    await self._attach()
.venv/lib/python3.11/site-packages/neo4j/_async/work/result.py:441: in _attach
    await self._connection.fetch_message()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:204: in inner
    await coroutine_func(*args, **kwargs)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:865: in fetch_message
    tag, fields = await self.inbox.pop(
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:77: in pop
    await self._buffer_one_chunk()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:54: in _buffer_one_chunk
    await receive_into_buffer(self._socket, self._buffer, 2)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:340: in receive_into_buffer
    n = await sock.recv_into(
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:221: in
recv_into
    res = await self._wait_for_read(self._reader.read, nbytes)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:173: in
_wait_for_io
    return await wait_for(io_fut, timeout)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait_for.py:118: in wait_for
    return fut.result()
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/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694: in
read
    await self._wait_for_data('read')

```

```

-----

self = <StreamReader transport=<_SelectorSocketTransport fd=10 read=polling write=<idle,
bufsize=0>>>
func_name = 'read'

```

```

async def _wait_for_data(self, func_name):
    """Wait until feed_data() or feed_eof() is called.

    If stream was paused, automatically resume it.
    """
    # StreamReader uses a future to link the protocol feed_data() method
    # to a read coroutine. Running two read coroutines at the same time

```

```

# would have an unexpected behaviour. It would not possible to know
# which coroutine would get the next data.
if self._waiter is not None:
    raise RuntimeError(
        f'{func_name}() called while another coroutine is '
        f'already waiting for incoming data')

assert not self._eof, '_wait_for_data after EOF'

# Waiting for data while paused will make deadlock, so prevent it.
# This is essential for readexactly(n) for case when n > self._limit.
if self._paused:
    self._paused = False
    self._transport.resume_reading()

self._waiter = self._loop.create_future()
try:
>     await self._waiter
E     RuntimeError: Task <Task pending name='Task-44' coro=<StreamReader.read()
running at
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>
cb=[_release_waiter(<Future pending...ask_wakeup()>)]() at /Users/akhilarya/UW Madison &
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait
_for.py:38]> got Future <Future pending> attached to a different loop

/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:
RuntimeError

```

During handling of the above exception, another exception occurred:

```
self = <neo4j._async.work.session.AsyncSession object at 0x120decd90>
```

```
@AsyncNonConcurrentMethodChecker._non_concurrent_method
```

```
async def close(self) -> None:
```

```
    """
```

```
    Close the session.
```

```
    This will release any borrowed resources, such as connections, and will
    roll back any outstanding transactions.
```

```
    """
```

```
    if self._closed:
```

```
        return
```

```
    if self._connection:
```

```
        if self._auto_result and self._state_failed is False:
```

```

    try:
        await self._auto_result.consume()
        await self._update_bookmark(self._auto_result._bookmark)
    except Exception:
        # TODO: Investigate potential non graceful close states
        self._auto_result = None
        self._state_failed = True

    if self._transaction:
        if self._transaction._closed() is False:
            # roll back the transaction if it is not closed
            await self._transaction._rollback()
        self._transaction = None

    try:
        if self._connection:
            await self._connection.send_all()
>         await self._connection.fetch_all()

.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:216:
-----
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:883: in fetch_all
    detail_delta, summary_delta = await self.fetch_message()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:865: in fetch_message
    tag, fields = await self.inbox.pop()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:77: in pop
    await self._buffer_one_chunk()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:54: in _buffer_one_chunk
    await receive_into_buffer(self._socket, self._buffer, 2)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:340: in receive_into_buffer
    n = await sock.recv_into(
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:221: in
recv_into
    res = await self._wait_for_read(self._reader.read, nbytes)
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.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:173: in
_wait_for_io
    return await wait_for(io_fut, timeout)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait_for.py:118: in wait_for
    return fut.result()
    ^^^^^^^^^^^^^^^^^

```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694: in
read
```

```
    await self._wait_for_data('read')
```

```
-----
self = <StreamReader transport=<_SelectorSocketTransport fd=10 read=polling write=<idle,
bufsize=0>>>
func_name = 'read'
```

```
async def _wait_for_data(self, func_name):
    """Wait until feed_data() or feed_eof() is called.
```

```
    If stream was paused, automatically resume it.
    """
```

```
    # StreamReader uses a future to link the protocol feed_data() method
    # to a read coroutine. Running two read coroutines at the same time
    # would have an unexpected behaviour. It would not possible to know
    # which coroutine would get the next data.
```

```
    if self._waiter is not None:
        raise RuntimeError(
            f'{func_name}() called while another coroutine is '
            f'already waiting for incoming data')
```

```
    assert not self._eof, '_wait_for_data after EOF'
```

```
    # Waiting for data while paused will make deadlock, so prevent it.
    # This is essential for readexactly(n) for case when n > self._limit.
    if self._paused:
        self._paused = False
        self._transport.resume_reading()
```

```
    self._waiter = self._loop.create_future()
    try:
```

```
>         await self._waiter
```

```
E         RuntimeError: Task <Task pending name='Task-45' coro=<StreamReader.read()
running at
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>
cb=[_release_waiter(<Future pending...ask_wakeup())>]() at /Users/akhilarya/UW Madison &
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait
_for.py:38]> got Future <Future pending> attached to a different loop
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:
RuntimeError
```

```
self = <tests.integration.test_milestone1.TestGraphStore object at 0x11c6d3b90>
engine = <nexus.epistemic.engine.EpistemicEngine object at 0x11cf7b10>
```

tests/integration/test_milestone1.py:136:

```
nexus/observability/tracing.py:148: in wrapper
    return await func(*args, **kwargs)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAA

nexus/epistemic/engine.py:247: in add_node
    return await self._graph.upsert_node(node)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAA

nexus/observability/tracing.py:148: in wrapper
    return await func(*args, **kwargs)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAA

nexus/epistemic/graph_store.py:164: in upsert_node
    async with self._session() as session:
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/contextlib.py:222: in __aexit__
    await self.gen.athrow(typ, value, traceback)
nexus/epistemic/graph_store.py:129: in _session
    async with self._driver.session(database=self._database) as session:
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:122: in __aexit__
    await self.close()
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:227: in close
    await self._disconnect()
.venv/lib/python3.11/site-packages/neo4j/_async/work/session.py:137: in _disconnect
    return await super()._disconnect(sync=sync)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAA

.venv/lib/python3.11/site-packages/neo4j/_async/work/workspace.py:285: in _disconnect
    await self._pool.release(self._connection)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_pool.py:498: in release
    await connection.reset()
```

```
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt5.py:443: in reset
    await self.fetch_all()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:883: in fetch_all
    detail_delta, summary_delta = await self.fetch_message()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async/io/_bolt.py:865: in fetch_message
    tag, fields = await self.inbox.pop()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:77: in pop
    await self._buffer_one_chunk()
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:54: in _buffer_one_chunk
    await receive_into_buffer(self._socket, self._buffer, 2)
.venv/lib/python3.11/site-packages/neo4j/_async/io/_common.py:340: in receive_into_buffer
    n = await sock.recv_into(
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:221: in
recv_into
    res = await self._wait_for_read(self._reader.read, nbytes)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/network/_bolt_socket.py:173: in
_wait_for_io
    return await wait_for(io_fut, timeout)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait_for.py:118: in wait_for
    return fut.result()
    ^^^^^^^^^^^^^^^^^^
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694: in
read
    await self._wait_for_data('read')
-----

self = <StreamReader transport=<_SelectorSocketTransport fd=10 read=polling write=<idle,
bufsize=0>>>
func name = 'read'
```

```
f'{func_name}() called while another coroutine is '  
f'already waiting for incoming data')
```

```
assert not self._eof, '_wait_for_data after EOF'
```

```
# Waiting for data while paused will make deadlock, so prevent it.  
# This is essential for readexactly(n) for case when n > self._limit.  
if self._paused:  
    self._paused = False  
    self._transport.resume_reading()
```

```
self._waiter = self._loop.create_future()  
try:
```

```
> await self._waiter
```

```
E RuntimeError: Task <Task pending name='Task-47' coro=<StreamReader.read()  
running at
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:694>  
cb=[_release_waiter(<Future pending...ask_wakeup()>)]() at /Users/akhilarya/UW Madison &  
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/neo4j/_async_compat/shims/_wait  
_for.py:38]> got Future <Future pending> attached to a different loop
```

```
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/asyncio/streams.py:526:  
RuntimeError
```

```
----- Captured log setup -----
```

```
INFO neo4j.notifications:result.py:345 Received notification from DBMS server:  
<GqlStatusObject gql_status='03N42', status_description='CONSTRAINT entity_id_unique  
FOR (e:Entity) REQUIRE (e.id) IS UNIQUE` already exists.', position=None,  
raw_classification='SCHEMA', classification=<NotificationClassification.SCHEMA: 'SCHEMA'>,  
raw_severity='INFORMATION', severity=<NotificationSeverity.INFORMATION:  
'INFORMATION'>, diagnostic_record={'OPERATION': '', 'OPERATION_CODE': '0',  
'CURRENT_SCHEMA': '/', '_classification': 'SCHEMA', '_severity': 'INFORMATION'}> for query:  
'CREATE CONSTRAINT entity_id_unique IF NOT EXISTS FOR (n:Entity) REQUIRE n.id IS  
UNIQUE'
```

```
INFO neo4j.notifications:result.py:345 Received notification from DBMS server:  
<GqlStatusObject gql_status='03N42', status_description='CONSTRAINT entity_name_ns FOR  
(e:Entity) REQUIRE (e.name, e.namespace) IS UNIQUE` already exists.', position=None,  
raw_classification='SCHEMA', classification=<NotificationClassification.SCHEMA: 'SCHEMA'>,  
raw_severity='INFORMATION', severity=<NotificationSeverity.INFORMATION:  
'INFORMATION'>, diagnostic_record={'OPERATION': '', 'OPERATION_CODE': '0',  
'CURRENT_SCHEMA': '/', '_classification': 'SCHEMA', '_severity': 'INFORMATION'}> for query:  
'CREATE CONSTRAINT entity_name_ns IF NOT EXISTS FOR (n:Entity) REQUIRE (n.name,  
n.namespace) IS UNIQUE'
```

```
INFO neo4j.notifications:result.py:345 Received notification from DBMS server:  
<GqlStatusObject gql_status='03N42', status_description='FULLTEXT INDEX entity_name_ft
```

```

FOR (e:Entity) ON EACH [e.name, e.embedding_text]` already exists.', position=None,
raw_classification='SCHEMA', classification=<NotificationClassification.SCHEMA: 'SCHEMA'>,
raw_severity='INFORMATION', severity=<NotificationSeverity.INFORMATION:
'INFORMATION'>, diagnostic_record={'OPERATION': '', 'OPERATION_CODE': '0',
'CURRENT_SCHEMA': '/', '_classification': 'SCHEMA', '_severity': 'INFORMATION'}> for query:
'\n  CREATE FULLTEXT INDEX entity_name_ft IF NOT EXISTS\n  FOR (n:Entity) ON EACH
[n.name, n.embedding_text]\n  '
INFO    neo4j.notifications:result.py:345 Received notification from DBMS server:
<GqlStatusObject gql_status='03N42', status_description="RANGE INDEX entity_created_at
FOR (e:Entity) ON (e.created_at)` already exists.', position=None,
raw_classification='SCHEMA', classification=<NotificationClassification.SCHEMA: 'SCHEMA'>,
raw_severity='INFORMATION', severity=<NotificationSeverity.INFORMATION:
'INFORMATION'>, diagnostic_record={'OPERATION': '', 'OPERATION_CODE': '0',
'CURRENT_SCHEMA': '/', '_classification': 'SCHEMA', '_severity': 'INFORMATION'}> for query:
'CREATE INDEX entity_created_at IF NOT EXISTS FOR (n:Entity) ON (n.created_at)'
INFO    neo4j.notifications:result.py:345 Received notification from DBMS server:
<GqlStatusObject gql_status='03N42', status_description="RANGE INDEX entity_type FOR
(e:Entity) ON (e.type)` already exists.', position=None, raw_classification='SCHEMA',
classification=<NotificationClassification.SCHEMA: 'SCHEMA'>, raw_severity='INFORMATION',
severity=<NotificationSeverity.INFORMATION: 'INFORMATION'>,
diagnostic_record={'OPERATION': '', 'OPERATION_CODE': '0', 'CURRENT_SCHEMA': '/',
'_classification': 'SCHEMA', '_severity': 'INFORMATION'}> for query: 'CREATE INDEX
entity_type IF NOT EXISTS FOR (n:Entity) ON (n.type)'
INFO    neo4j.notifications:result.py:345 Received notification from DBMS server:
<GqlStatusObject gql_status='03N42', status_description="RANGE INDEX entity_namespace
FOR (e:Entity) ON (e.namespace)` already exists.', position=None,
raw_classification='SCHEMA', classification=<NotificationClassification.SCHEMA: 'SCHEMA'>,
raw_severity='INFORMATION', severity=<NotificationSeverity.INFORMATION:
'INFORMATION'>, diagnostic_record={'OPERATION': '', 'OPERATION_CODE': '0',
'CURRENT_SCHEMA': '/', '_classification': 'SCHEMA', '_severity': 'INFORMATION'}> for query:
'CREATE INDEX entity_namespace IF NOT EXISTS FOR (n:Entity) ON (n.namespace)'
INFO    sentence_transformers.base.model:model.py:183 No device provided, using mps
INFO    httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/modules.json
"HTTP/1.1 307 Temporary Redirect"
WARNING  huggingface_hub.utils._http:_http.py:904 Warning: You are sending unauthenticated
requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
INFO    httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a
243fdf4706b3f48f1d95db1a4f5529b4d41/modules.json "HTTP/1.1 200 OK"
INFO    httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/config_sentence_tr
ansformers.json "HTTP/1.1 307 Temporary Redirect"

```


INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/config_sentence_transformers.json "HTTP/1.1 200 OK"

INFO sentence_transformers.base.model:model.py:992 Loading SentenceTransformer model from sentence-transformers/all-MiniLM-L6-v2.

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/config_sentence_transformers.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/config_sentence_transformers.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/README.md "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/README.md "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/modules.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/modules.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/sentence_bert_config.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/sentence_bert_config.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/adapter_config.json "HTTP/1.1 404 Not Found"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/config.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/processor_config.json "HTTP/1.1 404 Not Found"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/preprocessor_config.json "HTTP/1.1 404 Not Found"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/video_preprocessor_config.json "HTTP/1.1 404 Not Found"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/preprocessor_config.json "HTTP/1.1 404 Not Found"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/tokenizer_config.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/tokenizer_config.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/config.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/config.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/config.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/tokenizer_config.json "HTTP/1.1 307 Temporary Redirect"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a243fdf4706b3f48f1d95db1a4f5529b4d41/tokenizer_config.json "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: GET
https://huggingface.co/api/models/sentence-transformers/all-MiniLM-L6-v2/tree/main/additional_chat_templates?recursive=false&expand=false "HTTP/1.1 404 Not Found"

INFO httpx:_client.py:1025 HTTP Request: GET
https://huggingface.co/api/models/sentence-transformers/all-MiniLM-L6-v2/tree/main?recursive=true&expand=false "HTTP/1.1 200 OK"

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2/resolve/main/1_Pooling/config.json "HTTP/1.1 307 Temporary Redirect"

```

INFO httpx:_client.py:1025 HTTP Request: HEAD
https://huggingface.co/api/resolve-cache/models/sentence-transformers/all-MiniLM-L6-v2/1110a
243fdf4706b3f48f1d95db1a4f5529b4d41/1_Pooling%2Fconfig.json "HTTP/1.1 200 OK"
INFO httpx:_client.py:1025 HTTP Request: GET
https://huggingface.co/api/models/sentence-transformers/all-MiniLM-L6-v2 "HTTP/1.1 200 OK"
INFO httpx:_client.py:1740 HTTP Request: GET http://localhost:6333/collections "HTTP/1.1
200 OK"
INFO httpx:_client.py:1025 HTTP Request: GET http://localhost:6333 "HTTP/1.1 200 OK"
===== warnings summary =====
.venv/lib/python3.11/site-packages/opentelemetry/util/_importlib_metadata.py:32
/Users/akhilarya/UW Madison &
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/opentelemetry/util/_importlib_meta
data.py:32: DeprecationWarning: SelectableGroups dict interface is deprecated. Use select.
    return EntryPoints(ep for group_eps in eps.values() for ep in group_eps)

tests/integration/test_milestone1.py::TestGraphStore::test_upsert_and_retrieve_node
/Users/akhilarya/UW Madison &
F1/Projects/Nexus/Nexus/.venv/lib/python3.11/site-packages/pytest_asyncio/plugin.py:916:
PytestDeprecationWarning: Overriding the "event_loop_policy" fixture is deprecated and will be
removed in a future version of pytest-asyncio. Use the "pytest_asyncio_loop_factories" hook to
customize event loop creation.
    warnings.warn(

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== short test summary info =====
FAILED tests/integration/test_milestone1.py::TestGraphStore::test_upsert_and_retrieve_node -
RuntimeError: Task <Task pending name='Task-47' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestGraphStore::test_find_nodes_by_name -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestGraphStore::test_find_nodes_by_type -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestGraphStore::test_find_nodes_by_namespace
- RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestGraphStore::test_graph_stats - RuntimeError:
Task <Task pending name='Task-82' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestGraphStore::test_multi_hop_subgraph -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestVectorStore::test_semantic_search_relevance
- RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...
ERROR tests/integration/test_milestone1.py::TestVectorStore::test_vector_store_stats -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...
ERROR
tests/integration/test_milestone1.py::TestContextRouter::test_retrieve_returns_epistemic_context -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

```

ERROR

tests/integration/test_milestone1.py::TestContextRouter::test_retrieve_finds_graph_nodes -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

ERROR

tests/integration/test_milestone1.py::TestContextRouter::test_retrieve_finds_vector_memories -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

ERROR

tests/integration/test_milestone1.py::TestContextRouter::test_retrieve_latency_under_500ms -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

ERROR

tests/integration/test_milestone1.py::TestContextRouter::test_to_prompt_string_is_non_empty -
RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

ERROR

tests/integration/test_milestone1.py::TestMilestone1AcceptanceCriteria::test_AC2_multi_hop_q
uery - RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

ERROR

tests/integration/test_milestone1.py::TestMilestone1AcceptanceCriteria::test_AC3_retrieval_late
ncy - RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

ERROR

tests/integration/test_milestone1.py::TestMilestone1AcceptanceCriteria::test_AC4_dual_layer_r
etrieval - RuntimeError: Task <Task pending name='Task-82' coro=<StreamRe...

===== 1 failed, 4 passed, 2 warnings, 15 errors in 14.61s =====

make: *** [test-m1] Error 1

(.venv) akhilarya@Akhils-MacBook-Air Nexus %